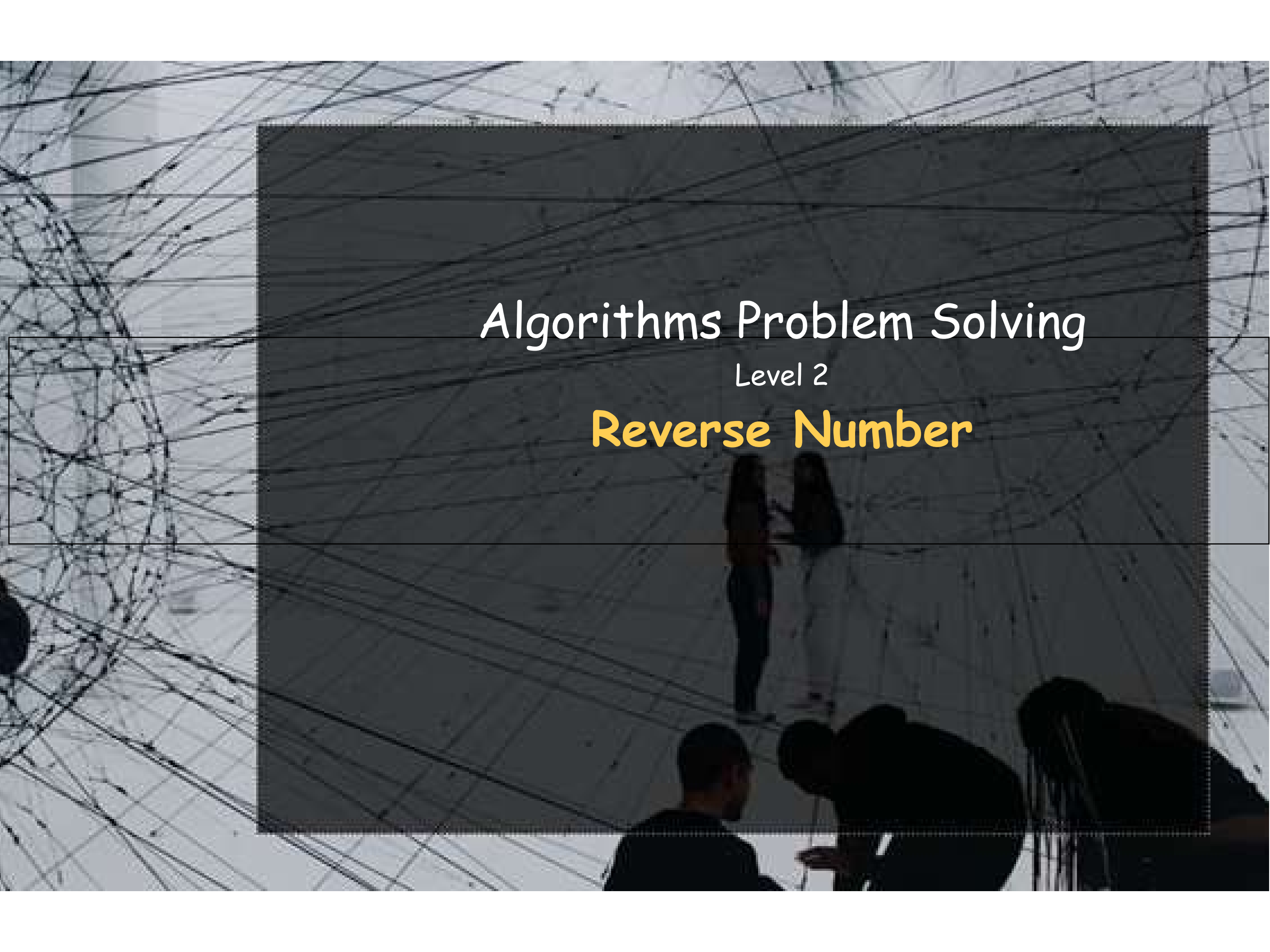


Algorithms Problem Solving

Level 2

Reverse Number



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Definition

- ◆ Reversing a number means changing the order of its digits from the last digit to the first digit.
- ◆ For example, `1234` becomes `4321`.

Explanation

- ◆ ``ReadNumber()`` reads a positive number from the user using a ``do-while`` loop.
- ◆ ``PrintResult()`` uses a ``while`` loop to process the number digit by digit.
- ◆ ``N % 10`` extracts the last digit of ``N``.
- ◆ ``N = N / 10`` removes the last digit from ``N``.
- ◆ ``cout << Remainder`` prints the extracted digit without moving to a new line.
- ◆ Because the digits are extracted from right to left, they are printed in reversed order.
- ◆ The program does not store the reversed number in another variable; it directly prints the digits in reverse order.

☐ Steps in main()

- ☐ Read a positive number N.
- ☐ Extract the last digit using `N % 10`.
- ☐ Remove the last digit using `N / 10`.
- ☐ Print the extracted digit.
- ☐ Repeat until N becomes `0`.

💡 Example

◆ Input:

``1234``

◆ ``1234 % 10 = 4`` → Print ``4``

◆ ``123 / 10`` → ``123``

◆ ``123 % 10 = 3`` → Print ``3``

◆ ``12 % 10 = 2`` → Print ``2``

◆ ``1 % 10 = 1`` → Print ``1``

➡ Result

``4321``

Note

- ◆ This program uses the same digit-extraction technique from the previous lessons.
- ◆ The important difference from `#5_Print_Digits_In_Reversed_Order` is that the digits are printed without `endl`, so they appear together as one reversed number.

⊘ Warning

- ◆ The variable `Sum` is declared but never used in this program.
- ◆ The program prints the reversed digits, but it does not actually create a reversed integer value that could be used later in calculations.



Comparison

Print Digits In Reversed Order

✓ `1234 → 4 3 2 1`

✓ Digits are usually printed separately.

Reverse Number

✓ `1234 → 4321`

✓ Digits are printed continuously to look like one reversed number.

✦ Summary

- ✓ `` % 10`` extracts the last digit.
- ✓ `` / 10`` removes the last digit.
- ✓ Digits are processed from right to left.
- ✓ ``cout << Remainder`` prints the digits without spaces or new lines.
- ✓ ``1234`` is displayed as ``4321``.

Thank you 😊